

Read $f(2)$ and the preimage of 2.5.

Rep. $f(2) = 0$; preimage of 2.5: -3 and 3.

Expression used for the route: f $(x) = 0{,}5x^2 - 2$.

5. Red thread · Method [ALG] Program: choose a number; multiply it by -2 ; add 10 ; divide by 2. Result for 7 ? Expression in x ?

Rep. $\dfrac{-14+10}{2} = -2$; in general $\dfrac{-2x+10}{2} = -x+5$.

Form 74 Mon. 31 May 2027 False positives and false negatives

1. Warming · Flash [PRB] Define "false positive" and "false negative".

Rep. False positive: positive test in a person who is not sick False negative: negative test in a person sick.

2. Heart · Method [PRB] 10 000 people, 2 % of patients; the test is positive for 90 % of patients and How many false negatives, false positives, total positives?

Rep. Illnesses 200: 180 positive, therefore 20 false negatives; non-malads 9,800: 980 false positives; positive: $180 + 980 = 1\,160$.

3. Heart + · Interpret [PRB] With the previous data, calculate the proportion of true patients among positive and comment.

Rep. $\dfrac{180}{1\,160} \approx 15.5\%$: The disease is rare, and most positives are false positives.

4. Return · True/false [PNP] True or false? Justify. The area of a square is proportional to the length of His side.

Rep. Wrong. $c = 1 \mapsto 1$ and $c = 2 \mapsto 4 \neq 2 \times 1$.

5. Red thread · Refuter [LIT] Conjecture: "for the integer n, $n^2 + n + 41$ is first." $n = 40$.

Rep. $40^2 + 40 + 41 = 1681 = 41^2$: The conjecture is false.

Successful tests prove nothing; a failed test refutes.

Consolidation, fluency and transition to the First

Form 75 Mar. 1 1st June 2027 Synthesis problems involving several areas

1. Warming · Flash [EQ] Solve 3 $(x-1) = x+5$.

Rep. $3x - 3 = x + 5$, $2x = 8$, $x = 4$.

2. Heart · Method [GEO] For what value of k points $A(1;1)$, $B(3;5)$ and $C(k;9)$ Are they aligned?

Rep. $\overrightarrow{AB}\begin{pmatrix}2\\4\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}k-1\\8\end{pmatrix}$: $2 \times 8 - 4(k-1) = 0$, therefore $k = 5$.

3. Heart + · Method [PRO] A price increases by 10% and then decreases by 10%. global?

Rep. $1{,}1 \times 0{,}9 = 0.99$: decrease of 1%.

4. Back · Flash [Sighs] Total area of a right pavement of dimensions 4, 3, 2 (cm) ?

Rep. $2(4 \times 3 + 4 \times 2 + 3 \times 2) = 52\ \text{cm}^2$.

5. Red thread · Flash [SOL] Calculate the exact volume of a 3 cm radius ball.

Rep. $V = \dfrac{4}{3}\pi \times 27 = 36\pi\ \text{cm}^3$.

Form 76 3 June 2027 Assessment of the six competences

1. Warming · Metacognition [MET] Citing the six mathematical skills worked in high school.

Rep. Search, model, represent, common difference / ratio, calculate, communicate.

2. Heart · Metacognition [MET] Associate: "to translate a statement by an equation"; "to plot a curve"; "justify equality by a written calculation" to a competency.

Rep. Model; represent; common difference / ratio (or communicate, depending on the emphasis on writing).

3. Heart + · Metacognition [MET] To solve 2 $x+3=11$, which competence is mainly mobilized?

Rep. Calculate (manipulate a first degree equation).

To be discussed orally: the equation of a problem is to "modelize".

4. Back · Problem [MOD] Water at 20 °C heats 3 °C per minute: $T(t)=20+3t$. At the end of How many minutes does it reach 65 °C?

Rep. $20+3t=65 \Leftrightarrow t=15$ min.

5. Red thread · Flash [EQ] Solve in $\mathbb{R}$: $0{,}5x+1{,}2=2{,}7$.

Rep. $x=3$

Form 77 Mon. 7 June 2027 Targeted Reactivation of Prerequisites for the First

1. Warming · Flash [DEV] Develop $(x+3)^2$.

Rep. x^2+6x+9.

2. Heart · Method [EQ] Verify that $x^2-5x+6=(x-2)(x-3)$ and deduce the solutions of $x^2-5x+6=0$.

Rep. $(x-2)(x-3)=x^2-5x+6$; zero product: $x=2$ or $x=3$.

3. Heart + · Anticipation [MOD] For $f(x)=x^2$, calculate the rate of increase $\dfrac{f(3)-f(1)}{3-1}$.

Rep. $\dfrac{9-1}{}=4$: prepares the concept of derived number in First.

Concept of First, simple reinvestment calculation.

4. Back · Construction programme [chuckles] We're setting M and N midpoints $[AB]$ and $[AC]$ And then we trace $[MN]$. That Can we say that (MN) and (BC) ? Calculate MN if $BC=9$ cm.

Rep. $(MN)\parallel(BC)$ and $MN=\dfrac{9}{2}=4.5$ cm.

5. Red thread · Write [DEM] Demonstrate that a triangle of sides 7, 24 and 25 is rectangle.

Rep. $7^2+24^2=49+576=625=25^2$: Reciprocity of Pythagoras

Form 78 March 8 June 2027 Synthesis issues: choosing and justifying a strategy

1. Warming · Code reading [ALG] What's sent back $f(2)$?

```
def f(x):
    return x**2 - 3*x + 1
```

Rep. $4-6+1=-1$.

2. Heart · Completion [ALG] Complete for function to return True where A, B, C are aligned:

```
def lines(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa)== ....
```

Rep. The determinant is compared to 0: == 0.

3. Heart + · Completion [ALG] Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        Return x
    Else:
        return ....
```

Rep. Return -x .